

Technical Document #393: Database Systems - Comprehensive Overview

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Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Relational databases organize data into tables with rows and columns.
- Sharding distributes data across multiple database servers.
- Database migration tools manage schema changes and data transformations.